

EEE 591 mini project4

Course: EEE 465/591 — Photovoltaic Energy Conversion

Mini Project #4: PV Systems — Stand-alone system analysis at Monument Valley, AZ (TMY: Page, AZ).

Part 1: Calculation of PV System Hourly Parameters

1) PV module energy (Wh) and current (Ah) for every hour of the year

Hourly PV output was computed using the TMY station for Page, AZ and a fixed mounting at 30° tilt, due south. The array consists of two 285 W modules (0.57 kW nameplate). The tilt-plane irradiance was calculated and propagated through hourly IV-curve evaluation. The resulting annual energy generated by the PV array is **1,657.37 kWh** and the corresponding total current is **34,528.63 Ah**. Figures 1a–1b show the hourly energy (Wh) and current (Ah) time series, respectively.

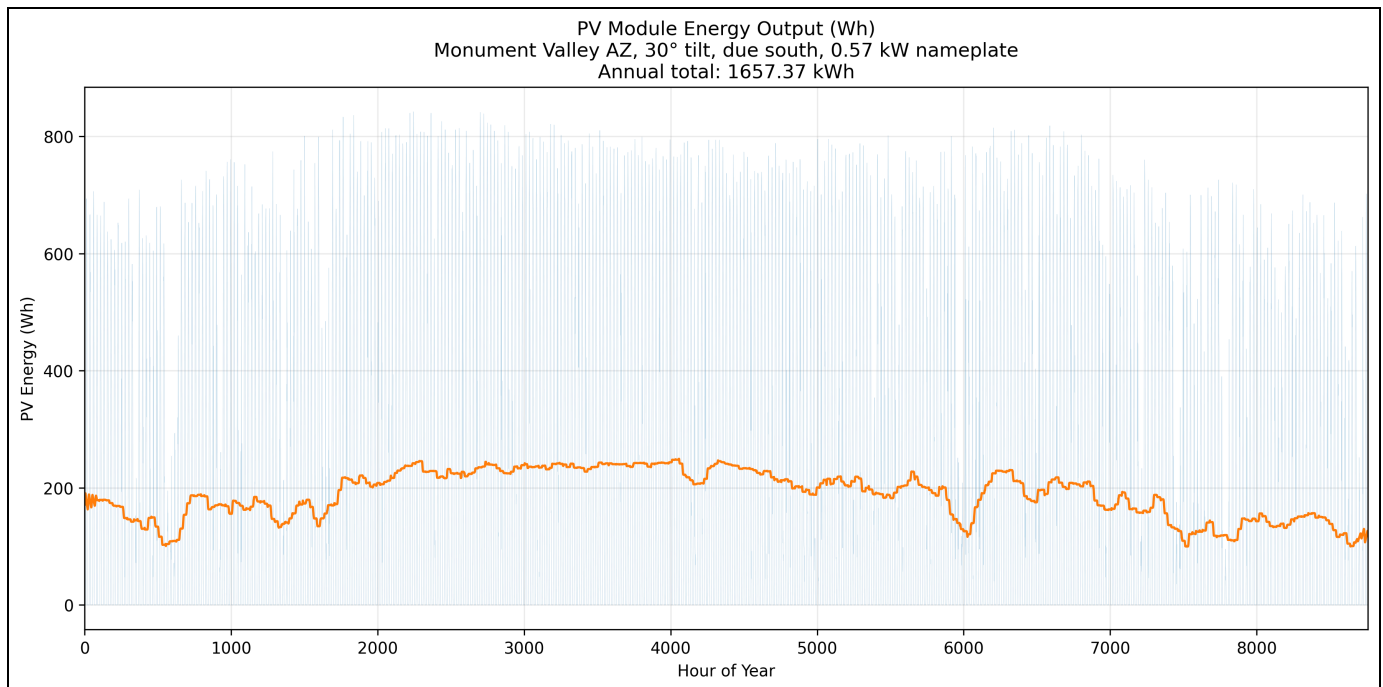


Figure 1a. PV Module Energy Output (Wh). Location: Monument Valley, AZ (TMY: Page, AZ). Mounting: 30° tilt, due south. Nameplate: two 285 W modules (0.57 kW). Annual total energy: 1,657.37 kWh.

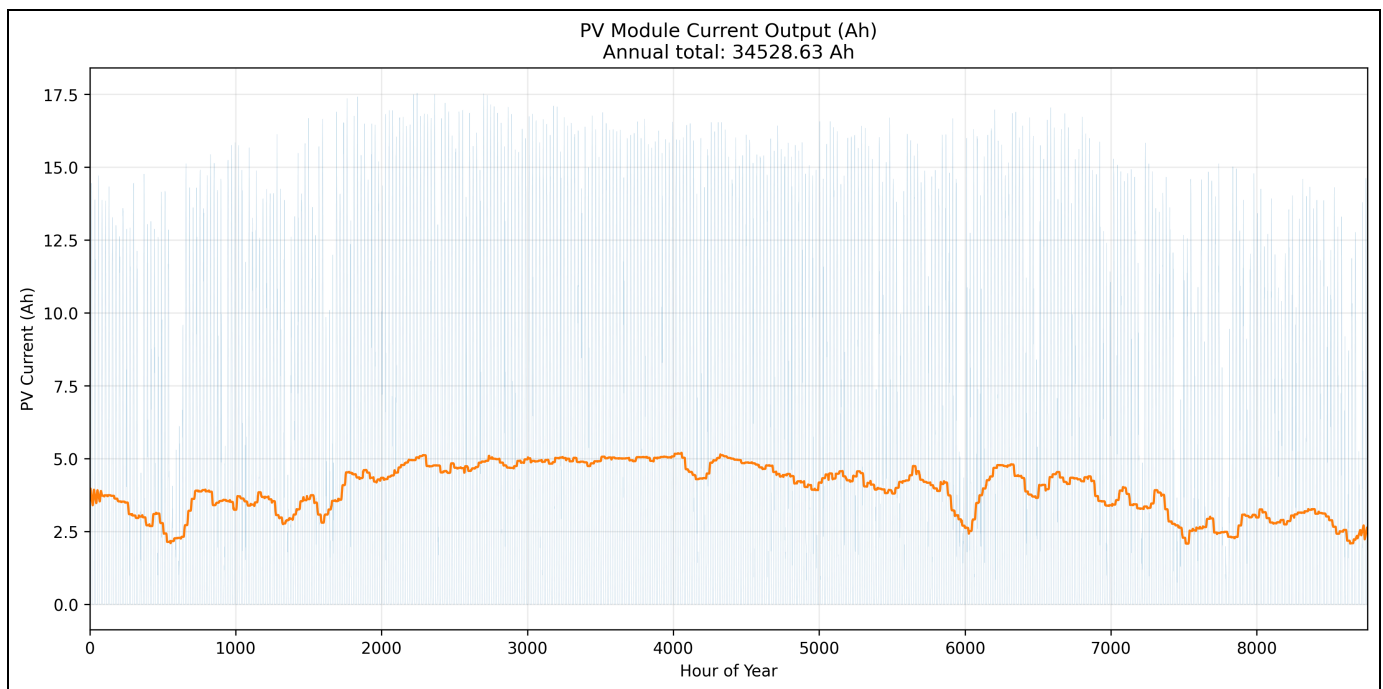


Figure 1b. PV Module Current Output (Ah). Annual total current: 34,528.63 Ah.

2) Hourly load profile with annual totals (kWh and Ah)

The load profile was constructed at one-hour resolution. DC lighting uses four 4 W LEDs (16 W total) from two hours before sunset until 22:00; an additional 24 W LED shop light operates from 19:00–22:00. Inverter-supplied loads include a 20 W screen and a 30 W laptop (both converted to DC equivalent at 95% inverter efficiency) operating from 19:00–22:00. A chest freezer of 95 W (converted to DC equivalent at 95% efficiency) runs with a 25% duty cycle (on one hour, off three hours). The total annual load is **335.04 kWh** and **6,979.99 Ah**. Figures 2a–2b show the stacked hourly load components (Wh) and the total load in Ah.

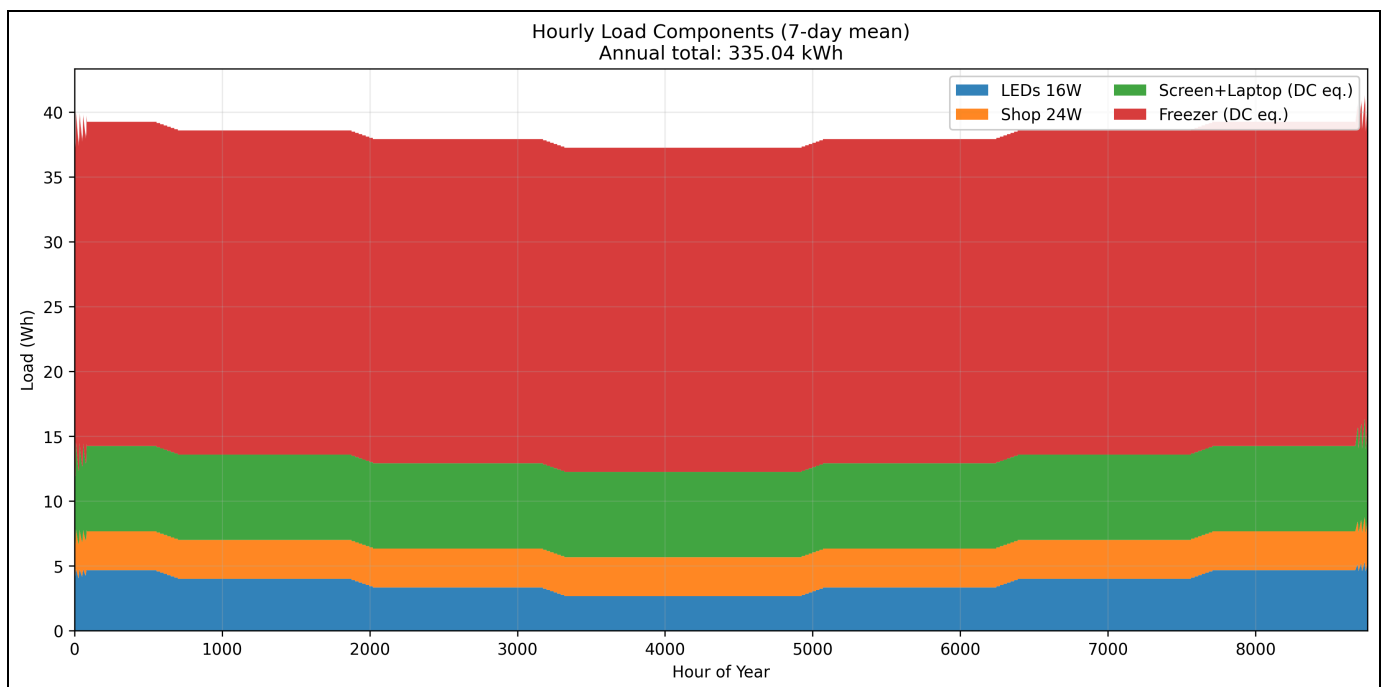


Figure 2a. Hourly Load Components (Wh), 7-day rolling mean displayed for readability. Annual total load energy: 335.04 kWh.

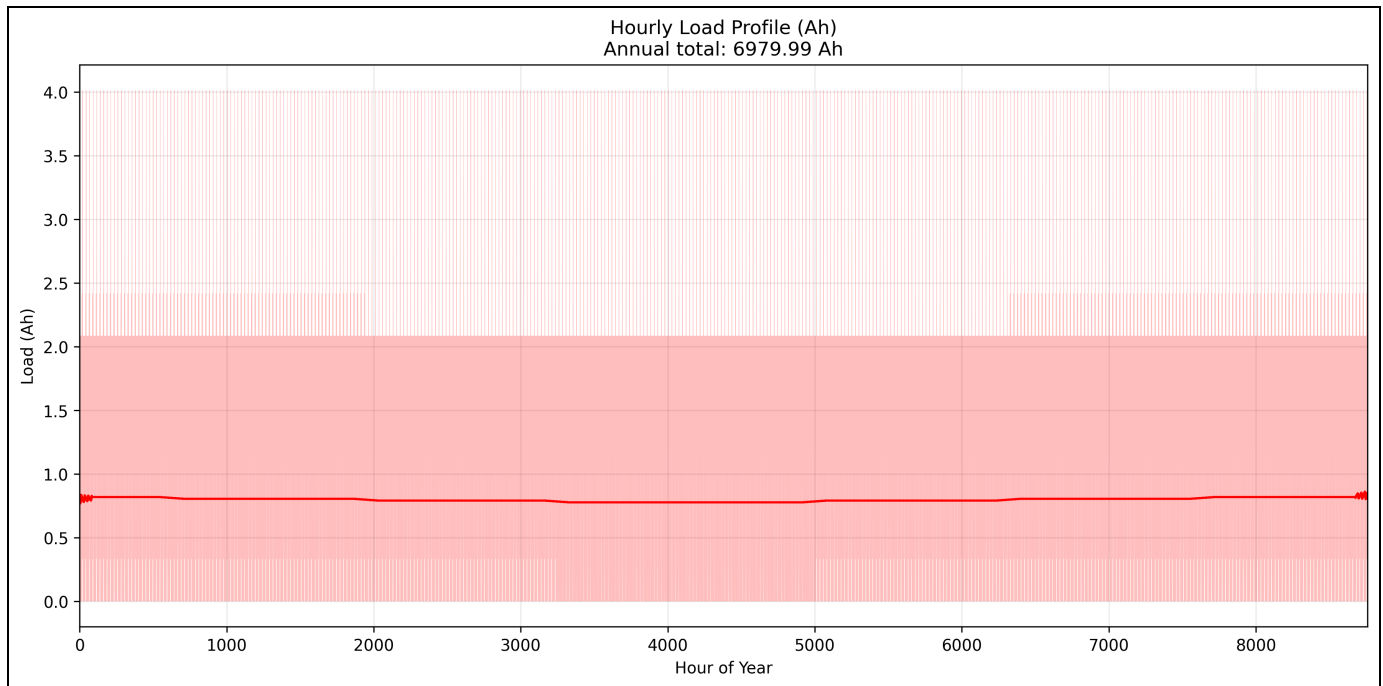


Figure 2b. Hourly Load Profile (Ah). Annual total load current: 6,979.99 Ah.

3) Hourly PV→Load, PV “dumped”, Battery SOC, Battery→Load, and Unavailable Load

With the battery initialized fully charged (48 V, 72 Ah), hourly energy routing was computed. The following cumulative flows were obtained: PV→Load **2,269.71 Ah**, PV→Battery **4,699.32 Ah**, PV Unused (dumped) **27,559.59 Ah**, Battery→Load **4,710.28 Ah**, and Unmet Load **0.00 Ah**. The Battery State of Charge (SOC) time series is plotted along with PV→Battery flow.

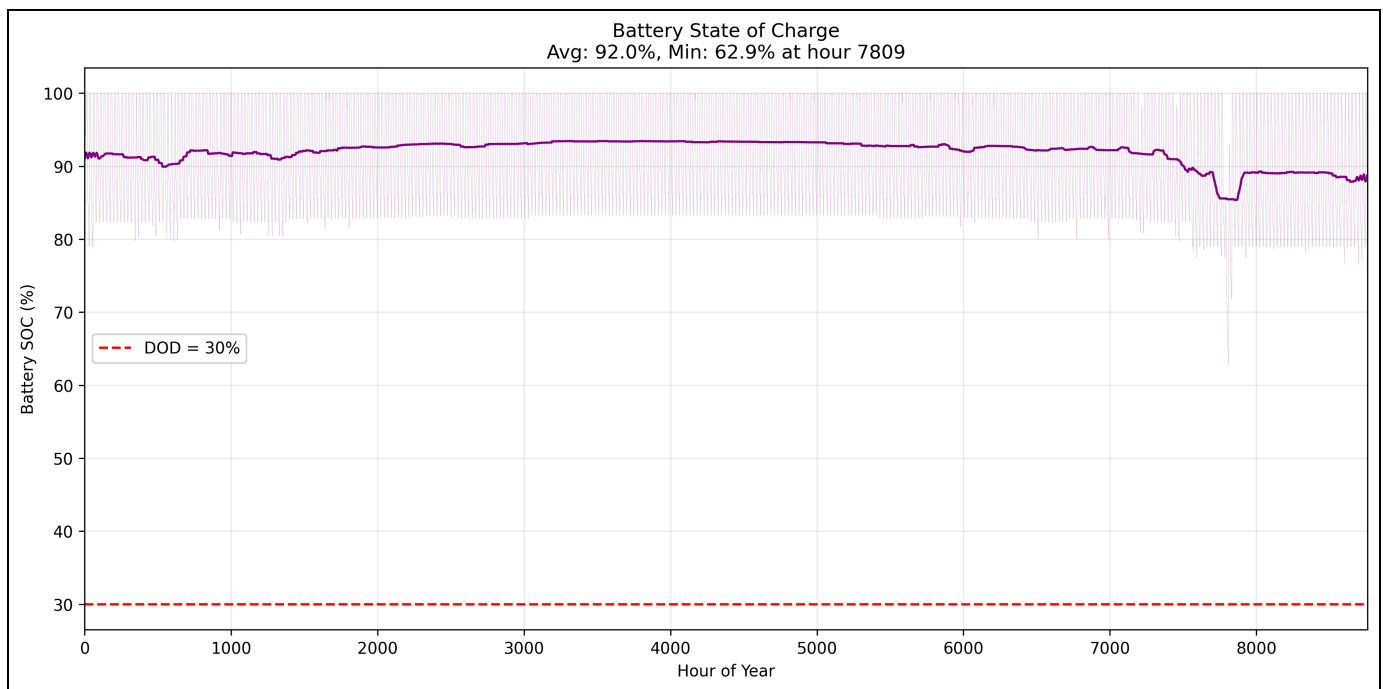


Figure 3a. Battery State of Charge (%). Average SOC: 91.95%. Minimum SOC: 62.88% at Hour 7809 (Day 326).

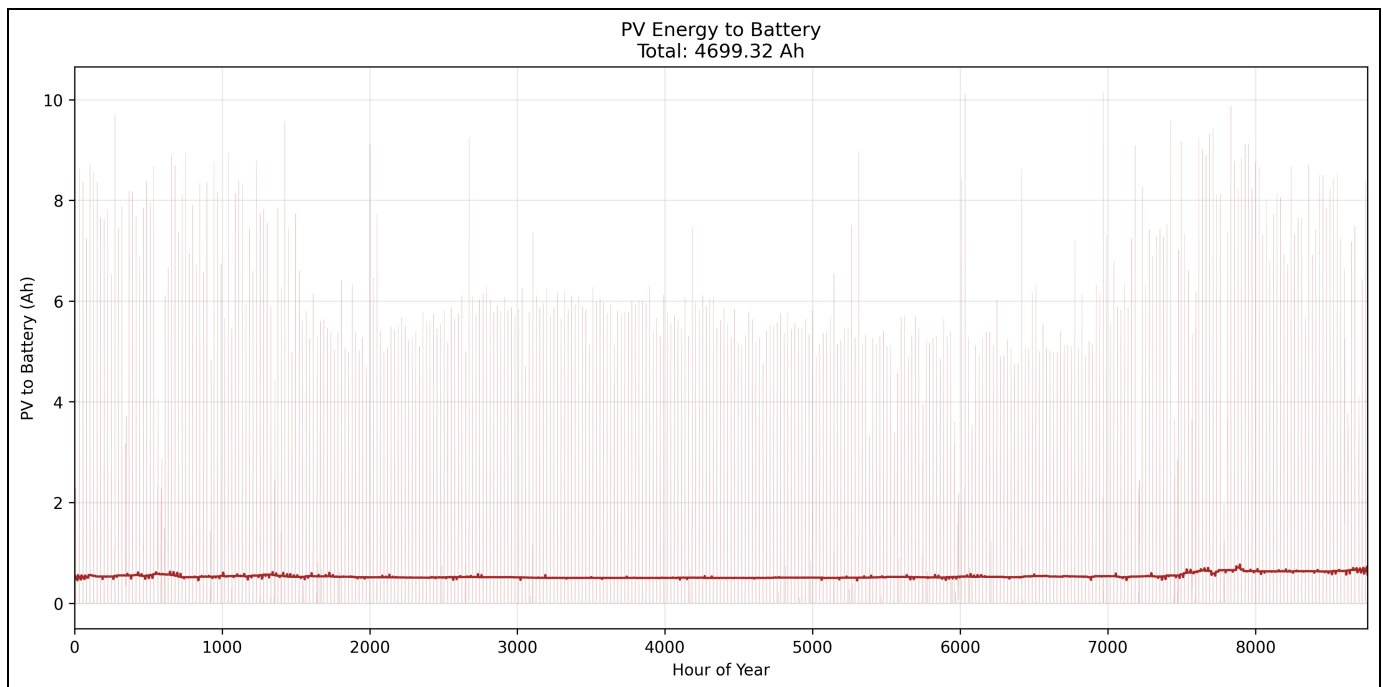


Figure 3b. PV Energy to Battery (Ah). Total PV→Battery: 4,699.32 Ah.

For clarity, the separate P/N/S flows (P = PV→Load, N = Not-served, S = Storage→Load) are shown in Figure 3c.

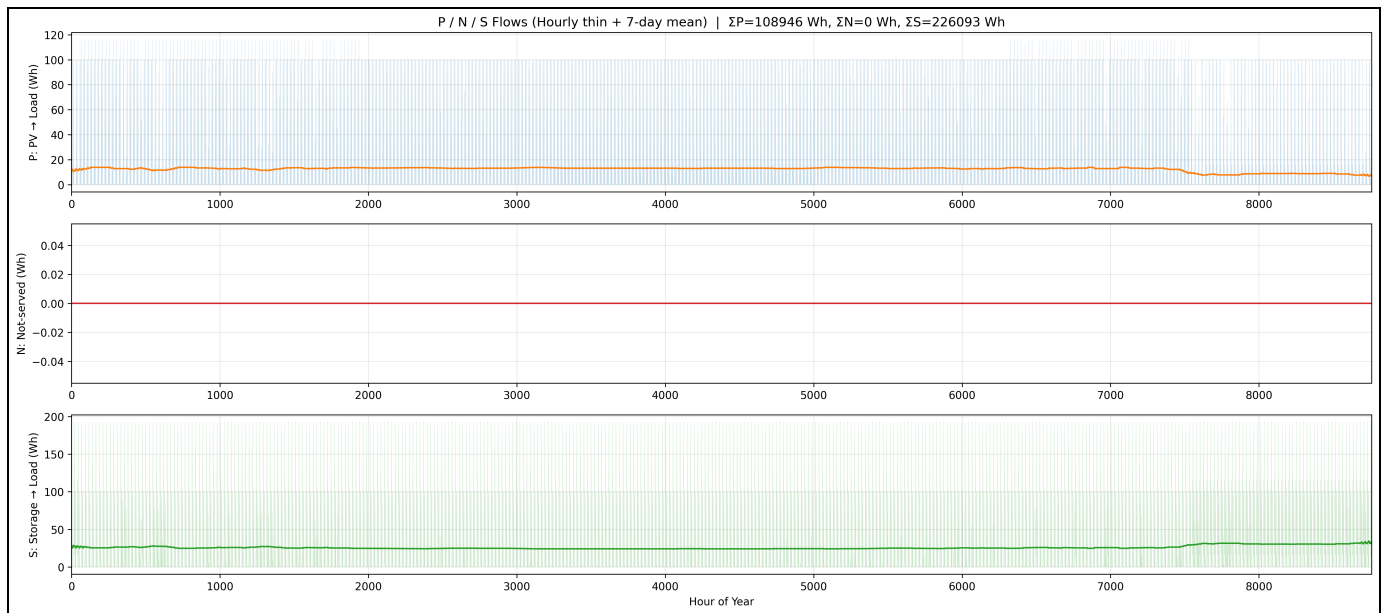


Figure 3c. P/N/S energy flows (hourly series plus 7-day mean). $\Sigma P = PV \rightarrow \text{Load}$; $\Sigma N = \text{Not-served}$; $\Sigma S = \text{Storage} \rightarrow \text{Load}$. Values used here: $P = 2,269.71 \text{ Ah}$ (convert to Wh at 48 V when needed), $N = 0.00 \text{ Ah}$, $S = 4,710.28 \text{ Ah}$.

Part 2: Analysis of PV System

1) Availability

Availability is defined as the ratio of the total powered load to the total requested load. The computed availability for the system is **100.00%**, indicating that all requested load was served.

2) Battery SOC statistics

Average SOC: 91.95%

Minimum SOC: 62.88% at Hour 7809 (Day 326)

3) Energy delivered to load vs. energy generated

The ratio of energy provided to the load to energy generated by the PV system is **0.2022** (computed consistently in Wh; identical ratio in Ah). For context, the fraction of PV energy used is **20.18%** and the fraction unused is **79.82%**.

Summary of computed results

- Annual PV energy: **1,657.37 kWh**; Annual PV current: **34,528.63 Ah**
- Annual load energy: **335.04 kWh**; Annual load current: **6,979.99 Ah**
- Availability: **100.00%**

- Avg SOC: **91.95%**; Min SOC: **62.88%** at Hour 7809 (Day 326)
- Energy-to-load / Energy-generated (Wh): **0.2022**
- PV→Load: **2,269.71 Ah**; PV→Battery: **4,699.32 Ah**; PV Unused: **27,559.59 Ah**
- Battery→Load: **4,710.28 Ah**; Unmet load: **0.00 Ah**